HITACHI

TEST PROCEDURE

Doc. No.	XEA-CP-004		
Revision	1	Page No.	P-1/37

PROJECT	East Delta Electricity Production Company El Ain El Sokhna Supercritical Thermal Power Plant Units 1&2 - 2x650 MW Gas/Oil Fired Units
	HITACHI Inspire the Next Sumitomo Corporation
CONTRACT NO. / SPECIFICATION NO.	CP-106
TITLE	COMMISSIONING PROCEDURE TURBINE TRIP INTERLOCK TEST

10056-9-V06-MUSG-00387 002
XEA-CP-004 01
Supplier Document Status
CODE: 4

Code 1 = Work may proceed
Code 2 = Revise and resubmit. Work may proceed subject to resolution of indicated comments.
Code 3 = Rejected. Revise and resubmit. Work may not Proceed.
Code 4 = Review not required.

Permission to proceed does not constitute acceptance or approval of design details, calculations, analyses, test methods or materials developed or selected by the Supplier and does not relieve the Supplier from full compliance with contractual obligations.

Responsible Engineer - Adel Eied
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Hitachi, Ltd.
Tokyo Japan
2013 -04 -15
Quality Assurance Division

Revision History

Rev. No.	Date	Revision Details	Prepared	Checked	Approved
0	2013.04.15	Original issue	—	—	—
1	2013.06.08	Revised in accordance with the comments	Y. ohara	A. E	D. Izumi

Purpose

- ■ Permission to Proceed
- ☐ Review
- ☐ Information
- ☐ Final

Distribution

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Thermal Power Plant
Testing & Commissioning Sec.

Date	2013.04.15
Prepared	Y.Ohara
Checked	A.Elhagry
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< ATTACHMENTS >

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TEST No.	DESCRIPTION OF TEST	RESULT
1.0	**SCOPE OF PROCEDURE** Turbine tripping system is the safety protection system for the boiler, turbine and generator against plant internal faults, auxiliary equipment failure and abnormal system conditions. Turbine trip interlock test is carried out for checking the proper function of the turbine tripping system in accordance with the interlock block diagram. This procedure applies to the Turbine Trip Interlock Test. (1) Main stop valve (A/B) (2) Control valve (A/B) (3) Combined reheat valve (A/B) (4) Ventilator valve (5) No.3～6,8 Extraction steam check valves	
2.0	**SYSTEM DESCRIPTION** Prior to the initial steaming to turbine, all turbine tripping interlock function and trip alarm shall be checked completely for the reliable operation. This procedure is stated with the tripping interlocks which can be checked on static condition. Therefore, the tripping test under running condition (I.e.overspeed test) is stated with a separate procedure. A real movement of the main valve (MSV, CV, CRV) are assumed to be the first test and last once for protection of the valve seat. At another test, confirm it in the opening and closeing situation of extraction steam check valves. Trip signal is carried out from the end of trip instrument. However, about impossible instrument ,trip signal is inputted near the trip instruments. (1) Trip check items 1) Master trip handle operated (Hydraulic fluid tank) 2) Turbine master trip PB (Control room) 3) Turbine master trip (TCS) (CRT OPRN) 4) HP turbine exhaust steam press very high (2 out of 3) 5) HP turbine exhaust steam temp very high (2 out of 3) 6) LP(A/B) exhaust hood press very high (condenser vacuum) (2 out of 3) 7) Main oil tank level very low (2 out of 3) 8) Hydraulic fluid tank level very low (2 out of 3) 9) Turbine trip oil press very low (2 out of 3) 10) Hydraulic fluid press very low (2 out of 3) 11) Bearing oil press very low (2 out of 3) 12) LP(A/B) exhaust hood temp very high (2 out of 3) 13) Turbine speed (turbine back-up overspeed) (2 out of 3) 14) Turbine speed (turbine overspeed) (2 out of 3)	

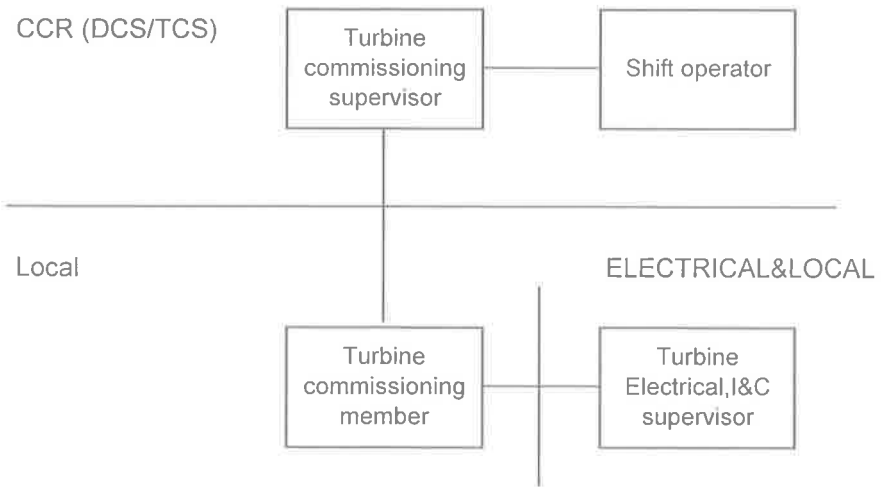
TEST No.	DESCRIPTION OF TEST	RESULT

15) Thrust bearing wear trip (front/rear) (2 out of 3)
16) Bearing shaft vibration very high and high (2 out of 3)
17) HP turbine first stage and exhaust steam difference press very high (2 out of 3)
18) Turbine shut down command (TCS) (ATS CRT OPRN)
19) Main steam temp very low (2 out of 3)
20) Owner's trip (2 out of 3)
21) TCS major trouble
22) Turbine trip command from electrical (A/B)
23) Boiler MFT (2 out of 3)
24) Gen collector housing difference temp very high (2 out of 3)
25) Gen stator cooling water inlet press very low (2 out of 3)
26) Gen stator cooling water outlet temp very high (2 out of 3)

(2) Test preparation
1) Steam turbine main valve adjustment shall be completed.
2) All supervisory instrument shall be adjusted and in-serviced.
3) Motor and air operated valves shall be adjusted and in-serviced.
4) Turbine reset/trip and main valve open/close shall be available.

< Remarks >
The first and the last once operate a main stop valve, control valve, combined reheat valve, ventilator valve, extraction steam check valves, master trip solenoid valve.
About others, operate only extraction steam check valves.
By the above, perform RSV test solenoid valve energize.
Turbine generator bearing vibration high high trip carries out only a combination of No.1 vibration and No.2 vibration.
(Turbine trip in No.1 vibration high high and No.2 vibration high high.)

3.0 **PERSONNEL ORGANIZATION**



TEST No.	DESCRIPTION OF TEST	RESULT
4.0	**REFERENCES** 1) Interlock block diagram (Steam turbine generator, and auxiliaries) 10056-9-V06-MUSG-00084 2) P&ID turbine lube oil & control oil system 10056-9-V06-MUSG-00099 ~ 00104 3) Diagram of steam turbine control mechanism 10056-9-V06-MUSG-00039 4) Turbine wiring diagram 10056-9-V06-MUSG-00111 5) Turbine protection logic 10056-9-V06-MUSG-00017 6) Generator protection/ breaker logic diagram (steam turbine generator) 10056-9-V06-MUSG-00139)	
5.0	**ATTACHMENTS** Commissioning forms 1) XEA-X1901-1 Master trip handle operated (Hydraulic fluid tank) 2) XEA-X1901-2 Turbine master trip PB (Control room) 3) XEA-X1901-3 Turbine master trip (TCS) (CRT OPRN) 4) XEA-X1901-4 HP turbine exhaust steam press very high 5) XEA-X1901-5 HP turbine exhaust steam temp very high 6) XEA-X1901-6 LP(A/B) exhaust hood press very high (condenser vacuum) 7) XEA-X1901-7 Main oil tank level very low 8) XEA-X1901-8 Hydraulic fluid tank level very low 9) XEA-X1901-9 Turbine trip oil press very low 10) XEA-X1901-10 Hydraulic fluid press very low 11) XEA-X1901-11 Bearing oil press very low 12) XEA-X1901-12 LP(A/B) exhaust hood temp very high 13) XEA-X1901-13 Turbine speed (turbine back-up overspeed) 14) XEA-X1901-14 Turbine speed (turbine overspeed) 15) XEA-X1901-15 Thrust bearing wear trip (front/rear) 16) XEA-X1901-16 Bearing shaft vibration very high and high 17) XEA-X1901-17 HP turbine first stage and exhaust steam difference press very high 18) XEA-X1901-18 Turbine shut down command (TCS) (ATS CRT OPRN) 19) XEA-X1901-19 Main steam temp very low 20) XEA-X1901-20 Owner's trip 21) XEA-X1901-21 TCS major trouble 22) XEA-X1901-22 Turbine trip command from electrical (A/B) 23) XEA-X1901-23 Boiler MFT 24) XEA-X1901-24 Gen collector housing difference temp very high 25) XEA-X1901-25 Gen stator cooling water inlet press low 26) XEA-X1901-26 Gen stator cooling water outlet temp high	

TEST No.	DESCRIPTION OF TEST	RESULT
6.0	**CHECK LIST (Preparation)** (1) The power supplies shall be available before trip test. 1) TCS & TSI control panel 2) Main oil pump (FL-MP-001A/B) 3) Vapor extractor (FL-MA-001A/B) 4) Hydraulic fluid pump (FJ-MP-001A/B) 5) No.3～6,8 Extraction steam check valves (AF-HV-680/660/650/640/630, AB-HV-621) 6) Ventilator valve (AB-HV-023) (2) Electrical, Instruments and control system are ready for turbine trip test. 1) Master trip solenoid valve A/B: AC-TY-606A/B 2) HP turbine exhaust steam press very high: AB-PIT-605AA/AB/B 3) HP turbine exhaust steam temp very high: AC-TT-622A/B/C 4) LP(A/B) exhaust hood press very high (condenser vacuum): AF-PIT-610AA/AB/AC/BA/BB/BC 5) Main oil tank level very low: FL-LIT-629A/B/C 6) Hydraulic fluid tank level very low: FL-LSLL-677A/B/C 7) Turbine trip oil press very low: FL-PSL-672A/B/C 8) Hydraulic fluid press very low: FL-PSLL-674A/B/C 9) Bearing oil press very low: FL-PSLL-623A/B/C 10) LP(A/B) exhaust hood temp very high: AC-TT-629AA/AB/AC/BA/BB/BC 11) Turbine speed (turbine back-up overspeed): AC-ST-666A/B/C 12) Turbine speed (turbine overspeed): AC-ST665A/B/C 13) Thrust bearing wear trip (front/rear): AC-ZT-664A/B/C 14) Bearing shaft vibration very high and high: AC-VT-650A/B ～655A/B, EY-VT-656A/B ～657A/B 15) HP turbine first stage and exhaust steam difference press very high: AC-PIT-613A/B/C, AB-PIT-605AA/AB/B 16) Main steam temp very low: AB-TT-601AA/AB/B 17) Gen collector housing difference temp very high: EY-TE-685A/B/C, 686A/B/C 18) Gen stator cooling water inlet press low: EY-PSL-618A/B/C 19) Gen stator cooling water outlet temp high: EY-TE-606A/B/C (3) For this commissioning, the following equipment and/or system are in service or standby. 1) MOP shall be running continuously. 2) HFP shall be running continuously. 3) Cooling water for oil coolers shall be available. 4) CWP shall be running continuously. 5) Gen SCW pump shall be running continuously. 6) Instrument air compresser shall be running continuously. 7) Master trip handle shall be reset position.	Date / Initial / / /

TEST No.	DESCRIPTION OF TEST	RESULT
6.0	(4) Test condition will be prepared as bellow. 1) LP(A/B) exhaust hood press very high (condenser vacuum) Simulate the LP(A/B) exhaust hood press normal by TCS. AFPIT610AA/AB/AC, AFPIT610BA/BB/BC 2) Electrical equipment trip Disconnect the incoming and outgoing trip signal of TCS Electrical equipment trip. 3) Boiler MFT trip Disconnect the incoming trip signal of TCS boiler MFT trip. 4) Boiler reheater protection Disconnect the outgoing trip signal of TCS Boiler reheater protection. 5) Owner's trip Disconnect the incoming trip signal of TCS Owner's trip. 6) Main steam temp very low Simulate the main steam temp by TCS. ABTT601AA, AB, B	/
7.0	**PROCEDURE** Test procedure details are shown Checklists and individual check sheet.	

XEA-X1901-1

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INTERLOCK TEST CHECK SHEET

No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
1	Master trip handle operated (Place: Hydraulic fluid tank)	1) Air supply to the extraction relay dump valve. 2) Turbine reset from TCS. 3) RSV: opened 4) MSV: opened (Simulate CV warming complete by TCS) 5) CV opened by CRT OPRN (CV: 10%opend, EL: 25%) 6) ICV: opened 7) Ventilator valve: closed 8) Extraction steam check valve cylinder: opened No.3 Extraction check valve No.4 Extraction check valve No.5P Extraction check valve No.5S Extraction check valve No.6 Extraction check valve No.8 Extraction check valve 9) Cold reheat steam check valve: opened	1) Turbine Tripped by hand trip valve (Hydraulic fluid tank)	1) TCS CRT Graphic (Drain valve) MSV(A),(B): Open → Close CV(A),(B) Position: about 10% → 0% RSV(A),(B): Open → Close ICV(A),(B): Open → Close Ventilator valve: Close → Open Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS CRT Graphic (HP/LP tb byp vlv) CR check valve: Open → Close 4) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

DATE OF TEST: _____
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INTERLOCK TEST CHECK SHEET						
No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
2	Turbine master trip PB (Place: Control room) **This test will check without valve stroking.**	1) Air supply to the extraction relay dump valve. 2) Turbine reset from TCS. 3) Extraction steam check valve cylinder: opened No.3 Extraction check valve No.4 Extraction check valve No.5P Extraction check valve No.5S Extraction check valve No.6 Extraction check valve No.8 Extraction check valve HITACHI	1) Push the Turbine master trip PB PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE MANUAL TRIP : TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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DATE OF TEST:
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INTERLOCK TEST CHECK SHEET						
No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
3	Turbine master trip (TCS) (CRT OPRN) (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Click the Turbine master trip (TCS) PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE MANUAL TRIP : TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

DATE OF TEST: _____
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INTERLOCK TEST CHECK SHEET

No	Name of test	Preparation before Test	Test Method	Confirmation Items / Operation Indication	Result	Remarks
4	HP turbine exhaust steam press very high (2 out of 3) Set point: > 70 barg (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the HP turbine exhaust steam press very high by TCS. (ABPIT605AA, AB, B)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : HP TURBINE EXHAUST STEAM PRESS VERY HIGH TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

DATE OF TEST:
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INTERLOCK TEST CHECK SHEET

No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
5	HP turbine exhaust steam temp very high (2 out of 3) Set point: > 420 degC (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the HP turbine exhaust steam temp very high by TCS. (ACTT622A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : HP TURBINE EXHAUST STEAM TEMP VERY HIGH TRIP : HP TURBINE EXHAUST STEAM TEMP HIGH	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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INTERLOCK TEST CHECK SHEET						
No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
6-1	LP(A) exhaust hood press very high (condenser vacuum) (2 out of 3) Set point: > 0.253 bar abs (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the LP(A) exhaust hood press very high by TCS. (AFPIT610AA, AB, AC)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : CONDENSER(A) VACUUM VERY LOW TRIP : CONDENSER(A) VACUUM LOW	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
6-2	LP(B) exhaust hood press very high (condenser vacuum) (2 out of 3) Set point: > 0.253 bar abs (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the LP(A) exhaust hood press very high by TCS. (AFPIT610BA, BB, BC)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : CONDENSER(B) VACUUM VERY LOW TRIP : CONDENSER(B) VACUUM LOW	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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INTERLOCK TEST CHECK SHEET						
No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
7	Main oil tank level very low (2 out of 3) Set point: < NOL-200mm (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Simulate the Main oil tank level very low by TCS. (FLLIT629A, B, C) PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : MAIN OIL TANK LEVEL VERY LOW TRIP : MAIN OIL TANK LEVEL LOW EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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DATE OF TEST: _____
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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
8	Hydraulic fluid tank level very low (2 out of 3) Set point: < NOL-180mm (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the Hydraulic fluid tank level very low by TCS. (FLLIT677A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : HYDRAULIC FLUID TANK LEVEL VERY LOW TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
9	Turbine trip oil press very low (2 out of 3) Set point: < 79 barg (Place: Hydraulic fluid tank) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Disconnect and jumper the trip signal cable from each pressure switch. FLPSL672A: cable No.TB10-1, 2 FLPSL672B: cable No.TB10-5, 6 FLPSL672C: cable No.TB10-9, 10	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
10	Hydraulic fluid press very low (2 out of 3) Set point: < 110 barg (Place: Hydraulic fluid tank) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Disconnect and jumper the trip signal cable from each pressure switch. FLPSL674A: cable No.TB9-1, 2 FLPSL674B: cable No.TB9-5, 6 FLPSL674C: cable No.TB9-9, 10 PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : HYDRAULIC FLUID PRESS VERY LOW TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
11	Bearing oil press very low (2 out of 3) Set point: < 0.69 barg (Place: Main oil tank) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Disconnect and jumper the trip signal cable from each pressure switch. FLPSLL623A: cable No.TBA-2,3 FLPSLL623B: cable No.TBA-8,9 FLPSLL623C: cable No.TBA-14,15	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : BEARING OIL PRESS VERY LOW TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
12-1	LP(A) exhaust hood temp very high (2 out of 3) Set point: < 107 degC (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the LP(A) exhaust hood temp very high by TCS. (ACTT629AA, AB, AC)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : LP(A) EXHAUST HOOD TEMP VERY HIGH TRIP : LP(A) EXHAUST HOOD TEMP HIGH		
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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
12-2	LP(B) exhaust hood temp very high (2 out of 3) Set point: < 107 degC (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the LP(A) exhaust hood temp very high by TCS. (ACTT629BA, BB, BC)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : LP(B) EXHAUST HOOD TEMP VERY HIGH TRIP : LP(B) EXHAUST HOOD TEMP HIGH		
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
13	Turbine speed (turbine back-up overspeed) (2 out of 3) Set point: <3345 ~ 3360 rpm (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the turbine speed (turbine back-up overspeed) by TCS. (ACST665A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : TURBINE BACK UP OVER SPEED TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
14	Turbine speed (turbine overspeed) (2 out of 3) Set point: <3315 ~ 3330 rpm (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the turbine speed (turbine overspeed) by TCS. (ACST666A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : TURBINE OVER SPEED TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		_____ HITACHI	_____ PGESCo	_____ EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
15	Thrust bearing wear trip (front/rear) (2 out of 3) Set point: > +0.8 mm < -1.28 mm (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Simulate the thrust bearing wear trip(1), (2), (3) by TCS. (TCA905, 906, 907) PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : THRUST BEARING WEAR TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
16	Bearing shaft vibration very high (2 out of 3) Set point: > 0.250mm (Very high) > 0.125mm (high) (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the bearing shaft vibration very high by TCS. (TCA905, 906, 907) **Simulate Turbine trip in No.1 vibration >0.250mm(Very high). And No.2 vibration >0.250mm (Very high).**	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : T-G BEARING VIBRATION VERY HIGH TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
17	HP turbine first stage and exhaust steam difference press very high (2 out of 3) Set point: Variable (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the generator load 35% 2) Simulate the HP turbine first stage and exhaust steam difference press very high by TCS. (ABPIT605AA, AB, B) (ACPIT613A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : HP TURBINE INLET/EXHAUST DIFF PRESS HIGH : HP TURBINE INLET/EXHAUST DIFF PRESS VERY HIGH TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

DATE OF TEST:
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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
18	Turbine shut down command (TCS) (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Push the Turbine shut down command (TCS) (ATS CRT OPRN)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP	_____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

DATE OF TEST:
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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
19	Main steam temp very low (2 out of 3) Set point: Variable (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Simulate the generator load 30% 2) Simulate the Main steam temp very low by TCS. (ABTT601AA, AB, B) (ACPIT613A, B, C) PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : MAIN STEAM TEMP VERY LOW TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

DATE OF TEST:
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INTERLOCK TEST CHECK SHEET

No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
20	Owner's trip (2 out of 3) (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the Owner's trip by TCS.	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP		
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
21	TCS major trouble (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the TCS major trouble by TCS.	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : TCS MAJOR TROUBLE : EHG MAJOR TROUBLE	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

XEA-X1901-22-1

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
22-1	Turbine trip command from electrical (A) (Place: GPP) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Making jumper on terminal block. (X01GA301-1, 2) "86-12" is activated.	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : TURBINE TRIP COMMAND FROM ELECTRICAL : GEN LOCKOUT RELAY (86-12) OPERATED	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

XEA-X1901-22-2

DATE OF TEST:
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INTERLOCK TEST CHECK SHEET						
No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
22-2	Turbine trip command from electrical (B) (Place: GPP) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Making jumper on terminal block. (X01GA301-3, 4) "86-22" is activated. PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : TURBINE TRIP COMMAND FROM ELECTRICAL : GEN LOCKOUT RELAY (86-22) OPERATED EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
23	Boiler MFT (2 out of 3) (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Simulate the Boiler MFT by TCS. PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : BOILER MFT EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
24	Gen collector housing difference temp very high (2 out of 3) Set point: > 45 degc (Place: Control room) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition)	1) Simulate the Gen collector housing difference temp very high by TCS. (EYTE685A, B, C) (EYTE686A, B, C)	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : GEN COLLECTOR RING DIFF TEMP VERY HIGH TRIP : GEN COLLECTOR RING DIFF TEMP HIGH : GEN COLLECTOR RING OUTLET AIR TEMP HIGH	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	
		HITACHI	PGESCo	EDEPC		

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
25	Gen stator cooling water inlet press very low (2 out of 3) Set point: < 0.9 barg (Place: Local panel) **This test will check without valve stroking.**	1) Test condition: same as the Test No.2 (Turbine Reset condition) HITACHI	1) Disconnect and jumper the trip signal cable from each pressure switch outlet terminal. (TB-2 No.1 and No.2) (TB-2 No.5 and No.6) Refer to DWG No. 310SC86-459 Sh.8 PGESCo	1) TCS CRT Graphic (Drain valve) Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : GEN STATOR INLET COOLING WATER PRESS VERY LOW TRIP EDEPC	_____ _____ _____ _____ _____ _____ _____ _____ _____ _____	

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No	Name of test	Preparation before Test	Test Method	Confirmation Items Operation Indication	Result	Remarks
26	Gen stator cooling water outlet temp very high (2 out of 3) Set point: > 72 degC (Place: Control room) **This test will check without valve stroking.**	1) Air supply to the extraction relay dump valve. 2) Turbine reset from TCS. 3) RSV: opened 4) MSV: opened (Simulate CV warming complete by TCS) 5) CV opened by CRT OPRN (CV: 10%opend, EL: 25%) 6) ICV: opened 7) Ventilator valve: closed 8) Extraction steam check valve cylinder: opened No.3 Extraction check valve No.4 Extraction check valve No.5P Extraction check valve No.5S Extraction check valve No.6 Extraction check valve No.8 Extraction check valve 9) Cold reheat steam check valve: opened HITACHI	1) Simulate the Gen stator cooling water outlet temp very high by TCS. (EYTE606A, B, C) PGESCo	1) TCS CRT Graphic (Drain valve) MSV(A),(B): Open → Close CV(A),(B) Position: about 10% → 0% RSV(A),(B): Open → Close ICV(A),(B): Open → Close Ventilator valve: Close → Open Turbine tripped: Green → Red 2) TCS CRT Graphic (Extract heaters) No.3 Ext. check valve: Open → Close No.4 Ext. check valve: Open → Close No.5P Ext. check valve: Open → Close No.5S Ext. check valve: Open → Close No.6 Ext. check valve: Open → Close No.8 Ext. check valve: Open → Close 3) TCS CRT Graphic (HP/LP tb byp vlv) CR check valve: Open → Close 4) TCS ANN: TURBINE TRIP : TURBINE TRIP OIL PRESS LOW TRIP : GEN STATOR OUTLET COOLING WATER TEMP VERY HIGH TRIP : GEN STATOR OUTLET COOLING WATER TEMP HIGH EDEPC		